~~OpenCV~~

Assignment No. 1

Computer vision Labratory

- Problem statement : In-stall opencv in python enviroment and perform the following operations :
  1. convert image from BGR to RGB
  2. Convert image to grayscale
  3. Display dimensions of image
  4. Convert RGB to HSV & HSV back to RGB
  5. Read, write & display image.

- Theory :

1. Explain all the OpenCV functions used in the experiment with proper description

Ans

(I) cv2.imRead()

Reads an image from specified path

cv2.imread(filename, flag)

filename → path of the image

flag → color mode

(II) cv2.cvtColor()

convert an image from one color space to another.

cv2.cvtColor(src, code)

eg. BGR2RGB

BGR2GRAY

RGB2HSV

HSV2RGB

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(iii) cv2.imshow()
cv2.imshow(window_name, image)
Display image in window

(iv) cv2.waitKey()
Wait for a key press for a specified time
cv2.waitKey(delay)
0 means wait indefinitely

(v) cv2.destroyAllWindow()
closes all openCV window

a. FAQ :

1. Classify the images.
Ans Digital image can be classified based on pixel value, color information & dimensions.

A. Binary images 0 or 1
represents black or white
Used in scanning document

B. Grayscale images
Pixel values range from 0 to 255
Represents intensity
Single channel image

C. Color images
Consists of multiple color channels.

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